

Environmental Sensor BLE Mesh

8-node BT SIG Mesh across ICAMS shop floor capturing temp, humidity, sound levels and feeding into grafana

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Role Project lead

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Program MMBSE

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Status Draft

Summary

A Bluetooth Low Energy Mesh (SIG) is being built to monitor environmental data across the ICAMS shop floor. The data that will be collected includes temperature, humidity, CO2, sound level, O2, particulate matter, and volatile organic compounds. The sensors used are the SCD40, DFRobot Sound Level Meter V1.0 and 2.0, SEN54, and DFRobot Oxygen Sensor.

The environmental sensors will be used to monitor working conditions and display on Grafana. Eventually, they will be connected to a ML algorithm to set thresholds that will trigger the smart switch system to power on and off dehumidifiers.

Background

This project will replace the previous system that was used to collect environmental sensor data through Bluetooth using the Arduino MKR Wifi 1010. This board was poorly suited for this task since Arduino does not have built-in support for BT SIG Mesh. Connections were limited to fewer than six nodes because of this, and data collection was spotty. Additionally, a new collector is being put in place and will run the script in a Docker container using Python, where all data will be pushed via MQTT through the broker Mosquitto.

Scope

In scope

- Provide data in a portable format
- Collect data
- Redundant nodes to ensure system always online
- Ensure ideal working conditions and monitoring

Out of scope

- ML Algorithm
- Control humidifiers
- Grafana display

Requirements

ID	Requirement	Priority
R1		-
R2	-	-

System design

Milestones

Phase	Deliverable	Target
1	-	-
2	-	-

Verification

Risks and open items

- **Item 1** needs resolution/decision
- **Item 2** needs resolution/decision

Approval

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Approver • Date